THE RESTRICTION OF A FLAT MODULE TO AN OBJECT OF THE SITE IS FLAT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites two related lemmas, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Exactness of extension by zero and its compatibility with tensor products. Stacks Project, Modules on Sites, Lemma lemma-extension-by-zero-exact (tag 09BG) and Lemma lemma-j-shriek-and-tensor (tag 09BH). Supplies the two tools that let exactness be transported to and from the site.

Source reference. Original lemma 03DJ: sources/stacks-chow/tags.txt, source lines 4422–4422. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0E8I: sources/stacks-chow/tags.txt, source lines 17818–17818. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0E8G: sources/stacks-chow/tags.txt, source lines 17816–17816. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let U be an object of $\mathcal{C}$. If $\mathcal{F}$ is a flat $\mathcal{O}$-module, then $\mathcal{F}|_U$ is a flat $\mathcal{O}_U$-module.*

Proof. Let $\mathcal{G}_1 \to \mathcal{G}_2 \to \mathcal{G}_3$ be an exact complex of $\mathcal{O}_U$-modules. Since $j_{U!}$ is exact (Lemma 03DJ) and $\mathcal{F}$ is flat as an $\mathcal{O}$-modules then we see that the complex made up of the modules

$$j_{U!}(\mathcal{G}_i \otimes_{\mathcal{O}_U} \mathcal{F}|_U) = j_{U!}\mathcal{G}_i \otimes_{\mathcal{O}} \mathcal{F}$$

(Lemma 0E8I) is exact. We conclude that $\mathcal{G}_1 \otimes_{\mathcal{O}_U} \mathcal{F}|_U \to \mathcal{G}_2 \otimes_{\mathcal{O}_U} \mathcal{F}|_U \to \mathcal{G}_3 \otimes_{\mathcal{O}_U} \mathcal{F}|_U$ is exact by Lemma 0E8G. □

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